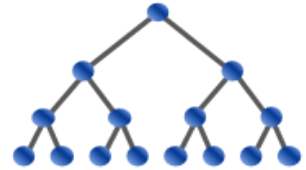


USA Computing Olympiad

[OVERVIEW](#)[DETAILS / FAQ](#)[TRAINING](#)[HISTORY](#)[RESOURCES](#)

USACO 2016 FEBRUARY CONTEST, BRONZE PROBLEM 2. CIRCULAR BARN

[Return to Problem List](#)

Contest has ended.

[Log in to allow submissions in analysis mode](#)

English (en)

Being a fan of contemporary architecture, Farmer John has built a new barn in the shape of a perfect circle. Inside, the barn consists of a ring of n rooms, numbered clockwise from $1 \dots n$ around the perimeter of the barn ($3 \leq n \leq 1,000$). Each room has doors to its two neighboring rooms, and also a door opening to the exterior of the barn.

Farmer John wants exactly r_i cows to end up in each room i ($1 \leq r_i \leq 100$). To herd the cows into the barn in an orderly fashion, he plans to unlock the exterior door of a single room, allowing the cows to enter through that door. Each cow then walks clockwise through the rooms until she reaches a suitable destination. Farmer John wants to unlock the exterior door that will cause his cows to collectively walk a minimum total amount of distance. Please determine the minimum total distance his cows will need to walk, if he chooses the best such door to unlock. The distance walked by a single cow is the number of interior doors through which she passes.

INPUT FORMAT (file cbarn.in):

The first line of input contains n . Each of the remaining n lines contain $r_1 \dots r_n$.

OUTPUT FORMAT (file cbarn.out):

Please write out the minimum total amount of distance the cows collectively need to travel.

SAMPLE INPUT:

```
5
4
7
8
6
4
```

SAMPLE OUTPUT:

```
48
```

In this example, the best solution is to let the cows enter through the door of the room that requires 7 cows.

Problem credits: Brian Dean

Contest has ended. No further submissions allowed.